ETHNOHACK 2026 · TRACK 2 · NATURAL PRODUCTS FOR DRUG DISCOVERY

Among hundreds of thousands of plant compounds, which deserve a closer look?

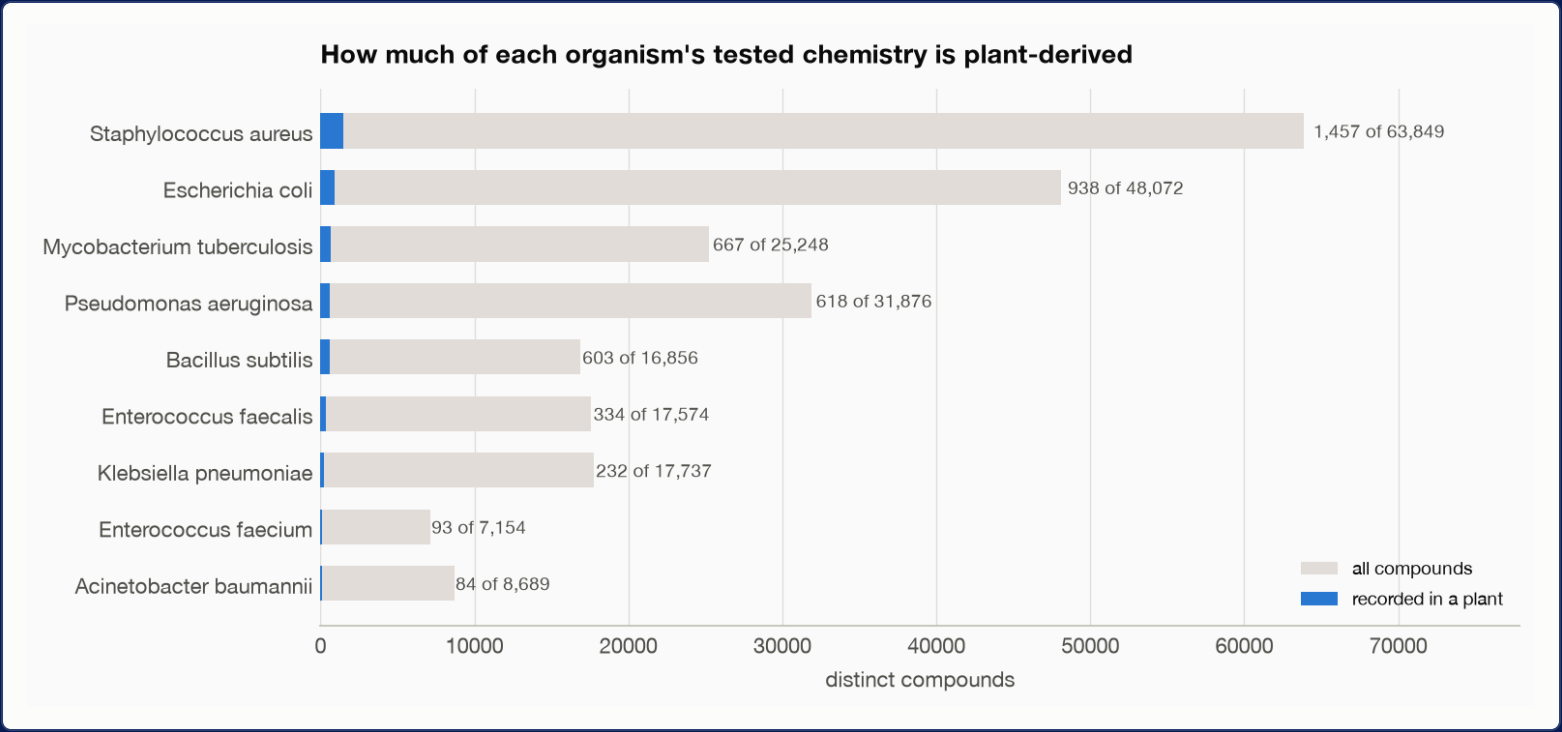
We took the track question literally: three bacterial pathogens, one model each, a ranked shortlist of plant-derived compounds for the bench, the decisions that built it, and a measured account of how far it can be trusted.

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The shortlists are live at `ethnohack-track2.vercel.app`

Everything shown is a model prediction over published laboratory measurements. Nothing here is a clinical claim, a statement of efficacy, or medical advice.

Medicine has measured a great deal of chemistry against these organisms. Almost none of it came from a plant.



ChEMBL MIC pull, before curation. After curation, *S. aureus* has 1,424 plant-derived compounds out of 58,191, 2.4% of them. Meanwhile 97,709 plant structures with a known source organism carry no measurement against any of these three at all.

You cannot look up an experiment nobody has run.

Three pathogens, three different cell envelopes, one model each.

We planned per-protein models. Then we measured the download, and changed the project to fit the data rather than the other way round.

99.6%

of MIC rows in our ChEMBL pull, across nine bacteria, are measured against the **whole organism**, not a single protein

35

single-protein targets carry any MIC at all, median 8 compounds, too few to train on

97,709

unmeasured plant structures screened, against 58,191 / 43,099 / 22,907 labelled compounds

THE THREE ENVELOPES

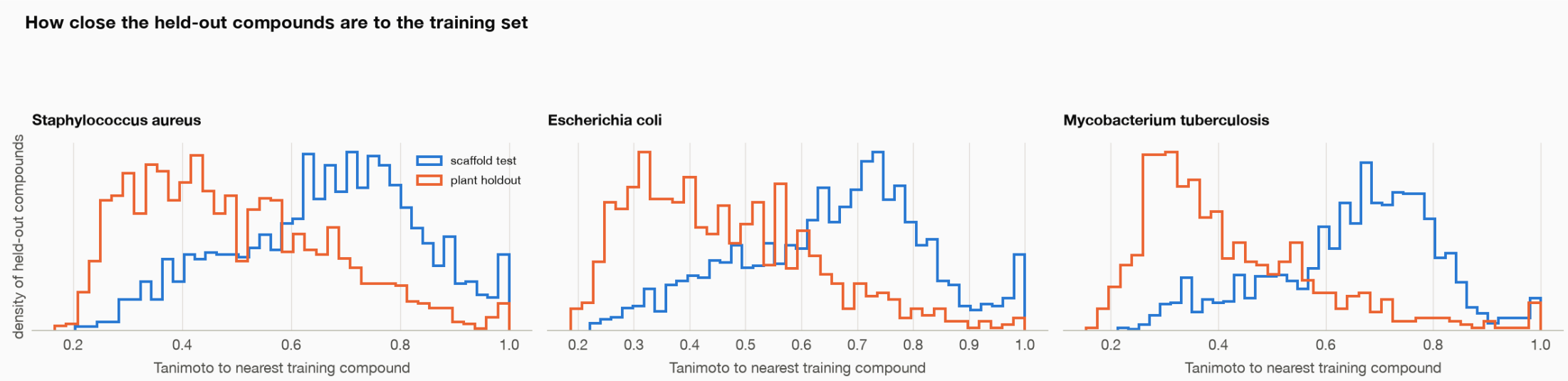
- *Staphylococcus aureus*: Gram-positive
- *Escherichia coli*: Gram-negative
- *Mycobacterium tuberculosis*: acid-fast, the hardest wall and the least data

WHAT THE MODEL THEREFORE CLAIMS

Whether a compound stops a given bacterium growing. Nothing about mechanism. The protein layer annotates a shortlist; it never ranks one.

Holding out chemical scaffolds is not enough. We hold out every plant-derived compound.

We cluster first and split the clusters, so no compound and no chemical scaffold appears on both sides of a split. The model cannot pass by having nearly seen the answer.



Removing shared ring skeletons removes none of the resemblance that matters. The median **scaffold-test** compound still has a close neighbour in training (*S. aureus*: `0.680`); the median **plant** compound sits far further out (`0.461`). Only `3.4%` of plant compounds have a near-twin, against `13.5%`.

That distance is why a nearest-training Tanimoto is printed beside every score we ship.

AUROC falls on the plant holdout for every pathogen. That is the number we rank on.

PATHOGEN	VALIDATION	SCAFFOLD TEST	PLANT HOLDOUT	95% CI	PLANT BASE RATE	P@10%	ENRICHMENT, TOP 5%
Staphylococcus aureus	0.880	0.863	0.719	0.689–0.749	0.253	0.570	2.45x
Escherichia coli	0.882	0.903	0.682	0.609–0.750	0.090	0.299	5.03x
Mycobacterium tuberculosis	0.846	0.815	0.629	0.582–0.677	0.237	0.435	2.29x

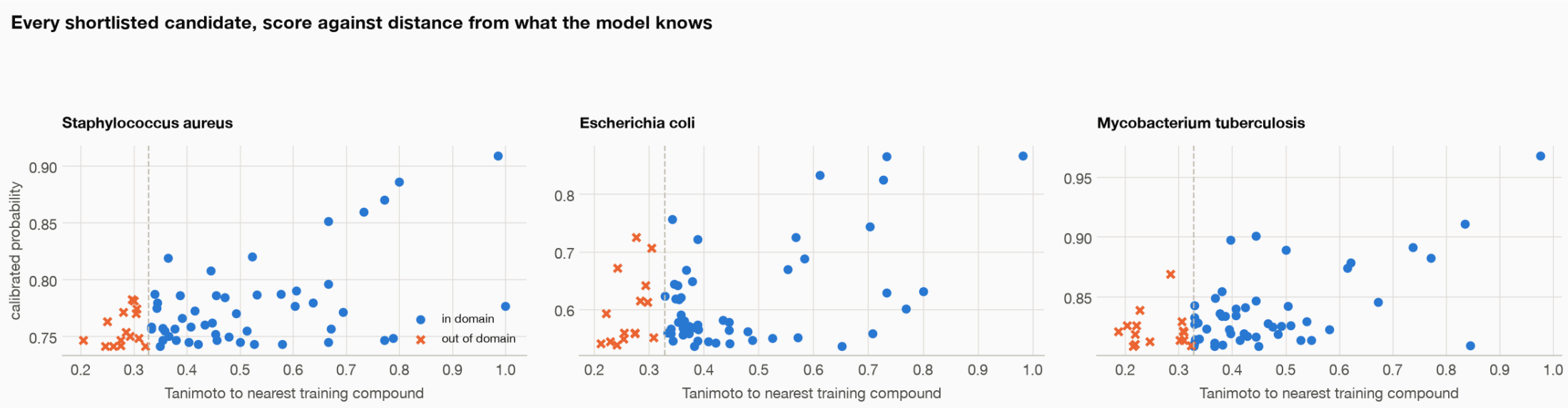
WHAT THIS MEANS FOR A LAB

Precision at the head of the ranking is what a chemist spends money against. Not a cure. A shorter queue for the bench.

ONE THING WE WILL NOT SMOOTH OVER

These rungs are **not** a tidy descending staircase. *E. coli* scores `higher` on the scaffold test (`0.903`) than on validation (`0.882`). The defensible claim is narrower: **the plant holdout is the hardest of the three, for every pathogen.**

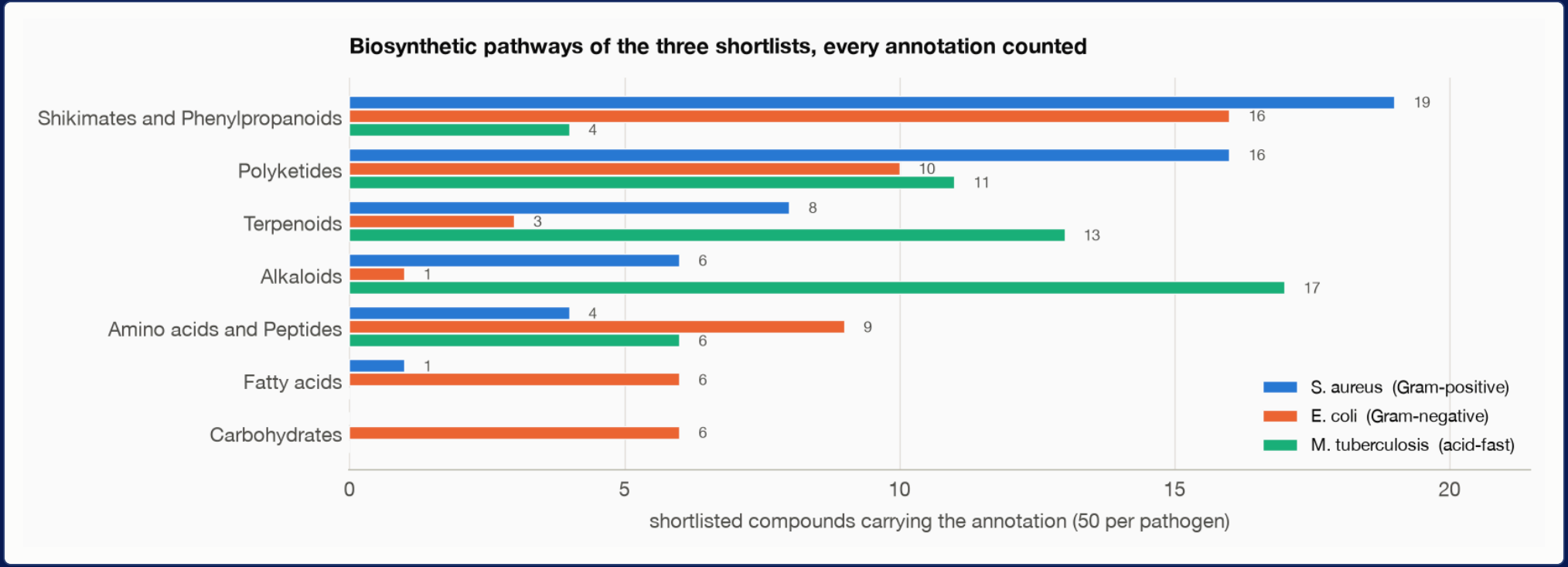
Every score travels with its distance from anything the model has seen.



The dashed line is the applicability domain at Tanimoto `0.328`: the 25th percentile of the plant holdout, the most distant set on which we actually measured the model. In red are the candidates we refuse to rank, published in a separate tier that states in its own words that its scores are not evidence of activity.

Per pathogen: 50 in-domain rows and 15 exploratory, each with predicted score and nearest-training Tanimoto, and with source plant, family and DOI wherever LOTUS records them. It does not for 12 / 8 / 3 of the 50, which slide 9 returns to.

The acid-fast shortlist is chemically distinct. The other two mostly are not.



M. tuberculosis, the only acid-fast organism here, takes the most alkaloids and terpenoids and the fewest shikimates. But *S. aureus* and *E. coli* overlap heavily, and **polyketides come out level across all three**.

> Nobody told any of these models about cell walls, and the acid-fast one saw fingerprints alone, with no physicochemical descriptors at all. A **hypothesis worth testing, not a finding**: that the wall a compound has to cross shapes which chemistry works against it. Only the acid-fast contrast is clean.

The thing we set out to show did not replicate. We are reporting it anyway.

The hypothesis: holding out whole plant **families** should be harder than holding out chemical **scaffolds**, because families share biosynthesis. We measured the gap four separate times, in a standalone experiment on 400–823 compounds, smaller than and separate from the main pipeline above.

MEASUREMENT	FAMILY – SCAFFOLD	PAIRED SD	FAMILY WORSE IN	VERDICT
S. aureus	−0.073	0.067	8 of 10	direction holds
E. coli	+0.004	0.139	2 of 10	reverses
M. tuberculosis	−0.097	0.158	7 of 10	spread exceeds effect
Earlier run, NPASS pull	−0.130	–	10 of 10	source since dropped

> The magnitude never settles, and on *E. coli* it reverses outright. Two of the three ChEMBL estimates rest on test sets of about eighty and a hundred compounds. **So it goes in as an honest negative, not as our contribution**, and the cluster-level scaffold split, which replicates cleanly on all three, is what carries the evaluation on the previous slides. Paired sds are the experiment's own output; `runs.jsonl` records the per-split ones.

The part we would rather a judge heard from us than found alone.

WHAT OUR OWN CHECKS CAUGHT

- The **top-ranked candidate in all three** is a near-copy of a training compound, at Tanimoto `0.986` / `0.982` / `0.976`. Two of the three are non-plant polyketides of the same class, an anthracycline and a tetracycline, with no source plant and no DOI.
- One row ships at Tanimoto `1.000`, correctly flagged as a fingerprint twin of a training compound.
- No source plant on 12 / 8 / 3 of 50 rows, and no plant family on 15 / 8 / 3. DOI coverage 38 / 42 / 47 of 50.
- Per-compound attributions came out numerically impossible, so they are withheld from the shortlists and the site rather than shipped untrusted. They remain in the evidence files, flagged.

LIMITS OF THE METHOD ITSELF

- Labels carry publication bias: most published MIC values sit at or below the active threshold, because inactive compounds are reported less often.
- No mechanism. A whole-organism readout cannot tell you what a compound binds.
- Model and shortlist stages have **not** been independently reproduced; curation and splits have.
- No experimental validation of any kind. Not one of these compounds has been put on a plate by us.

> A screen that only reports its wins is not a result.

Every model and evaluation number here is read from a committed result file.

57

curation and split metrics re-run end to end from the raw downloads on a second machine, all matched to the last decimal. The model and shortlist stages were not re-run this way

22

named pipeline runs, each writing its config hash and metrics to `results/runs.jsonl`

0

model or evaluation metrics typed by hand into the web app; it reads `results/` at build time and retrains nothing

DATA AND LICENCES

- **ChEMBL** (CC BY-SA): the MIC activity labels, whole-organism.
- **LOTUS** (CC0): the plant-to-compound edge, with references.
- **GBIF**: cross-check of which source organisms are plants.
- **NPASS** was evaluated and **dropped**: its MIC table is a copy of ChEMBL, so we cite ChEMBL directly.

WHAT IS NEXT

- The protein layer, from the IC50 and Ki download that already exists.
- **The honest next step is not a fourth pathogen.** It is putting ten of these on a real plate and publishing the hit rate next to the predicted one, including if it is worse.

> The plant records carry traditional knowledge. Anything built on this carries the Nagoya Protocol and Local Contexts with it as a condition, not a footnote. Generative-AI assistance is disclosed in the repository README.

All three shortlists, with every score, distance, source plant and DOI: `ethnohack-track2.vercel.app`